1.2 - Windows Basics

You have to know your Windows ABCs to even begin!

By Clement Chan

Applications

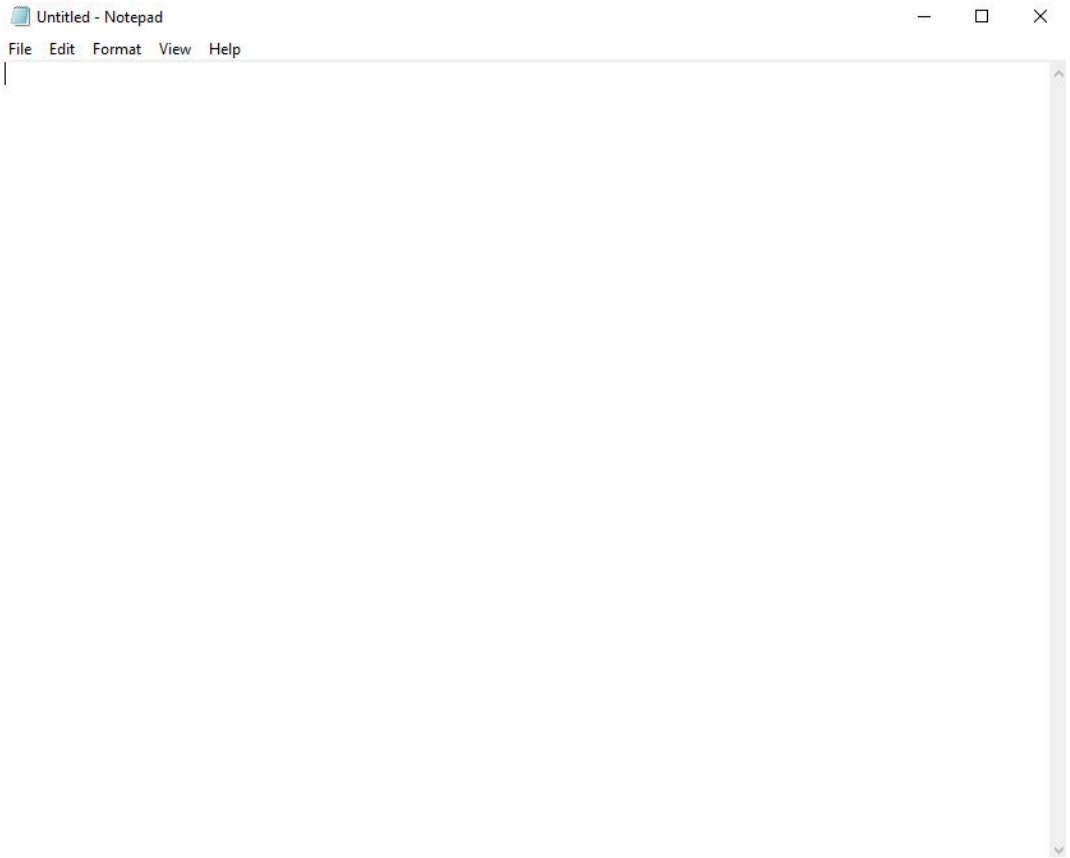
Applications

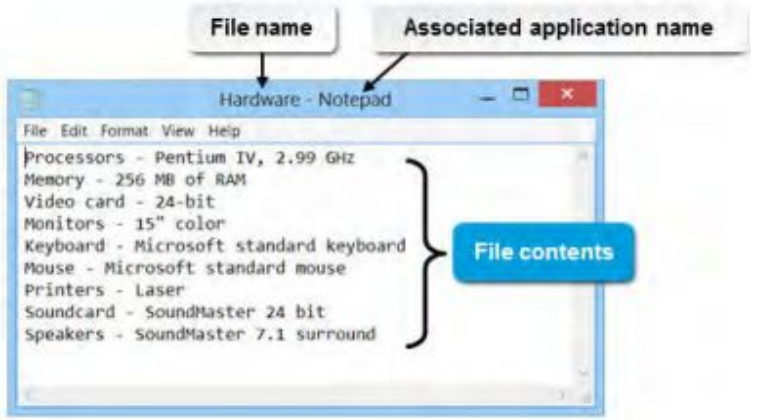
Programs that provides specific functionality

Microsoft Office products, Google Chrome, Steam, etc



Notepad - basic text editing software

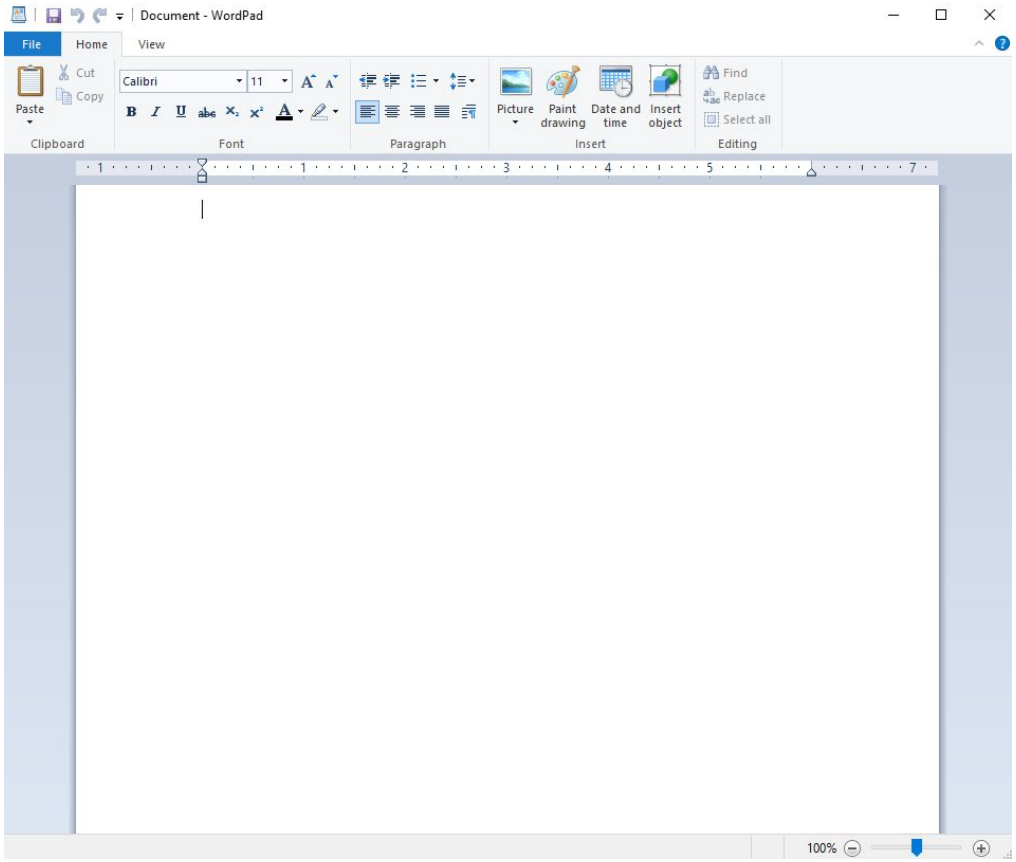




Simplistic design, comes with all Windows versions



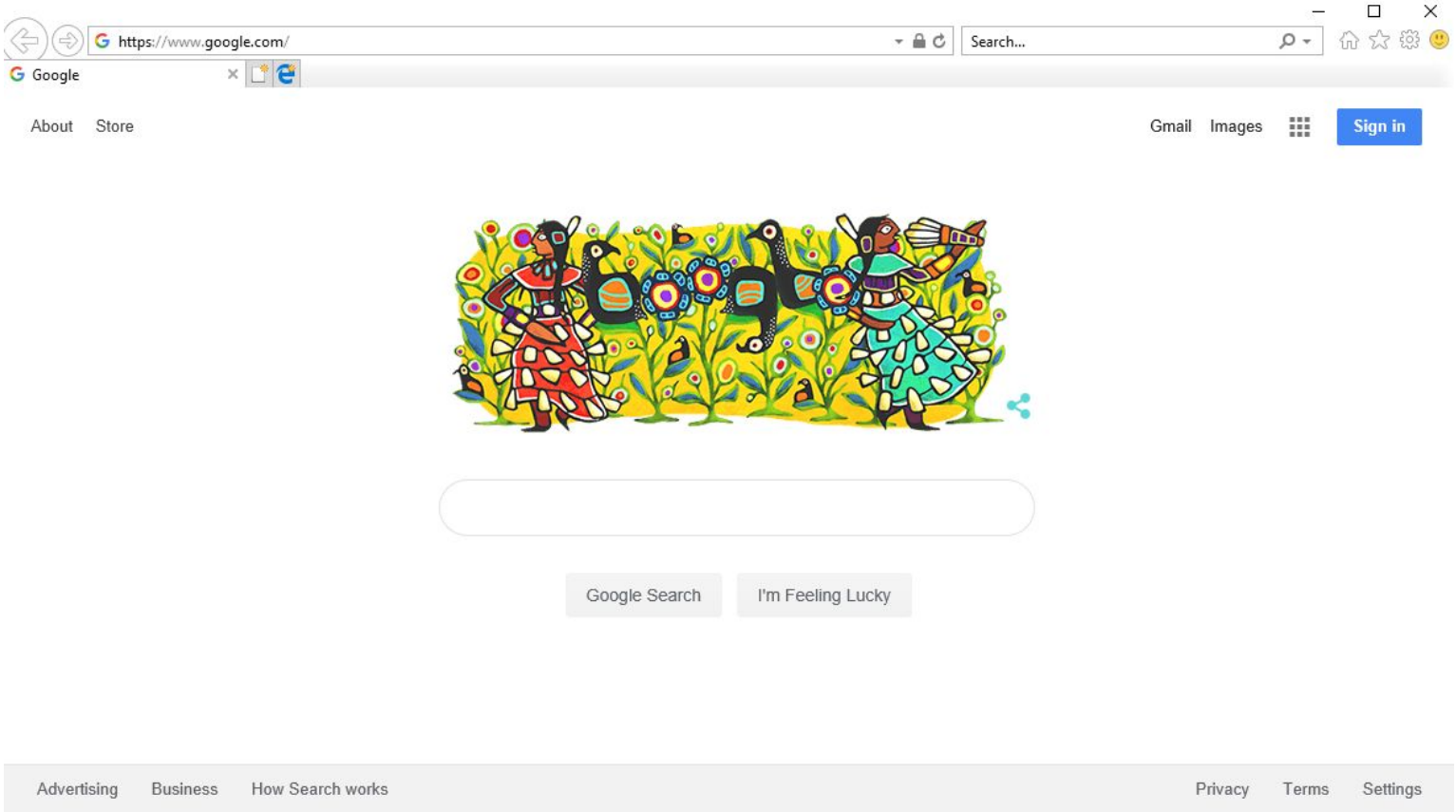
Wordpad - text editing software with formatting



Similar to Microsoft Office, preinstalled and free



Internet Explorer - web browser to access the Internet



Infamously slow, comes with Windows

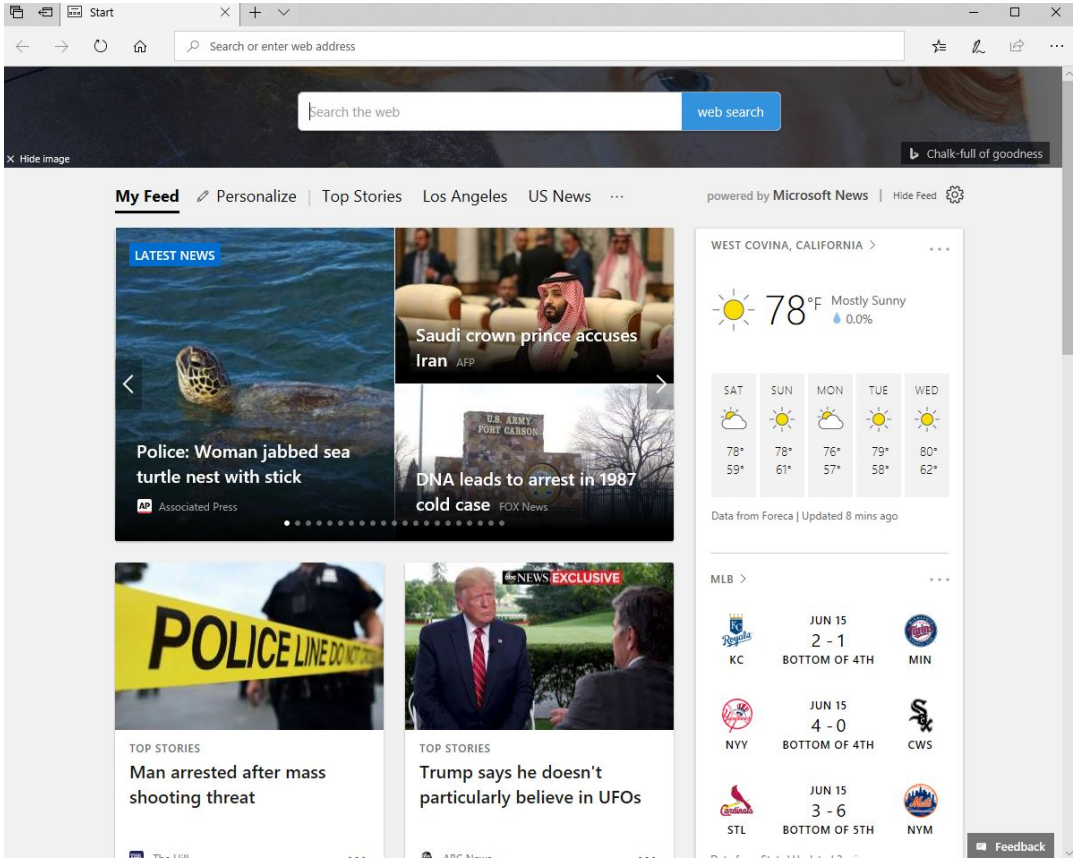
Microsoft
Edgy



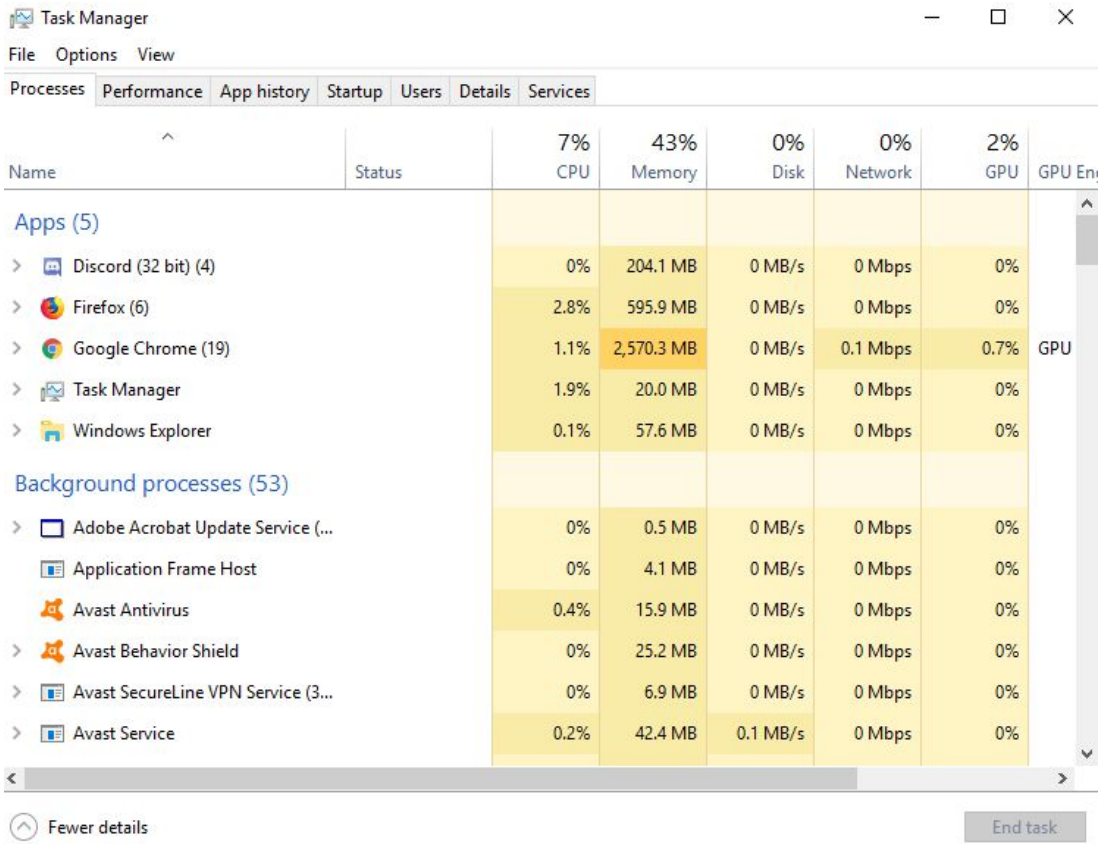
Microsoft Edgy - REEEEE



Microsoft Edge - better version of Internet Explorer



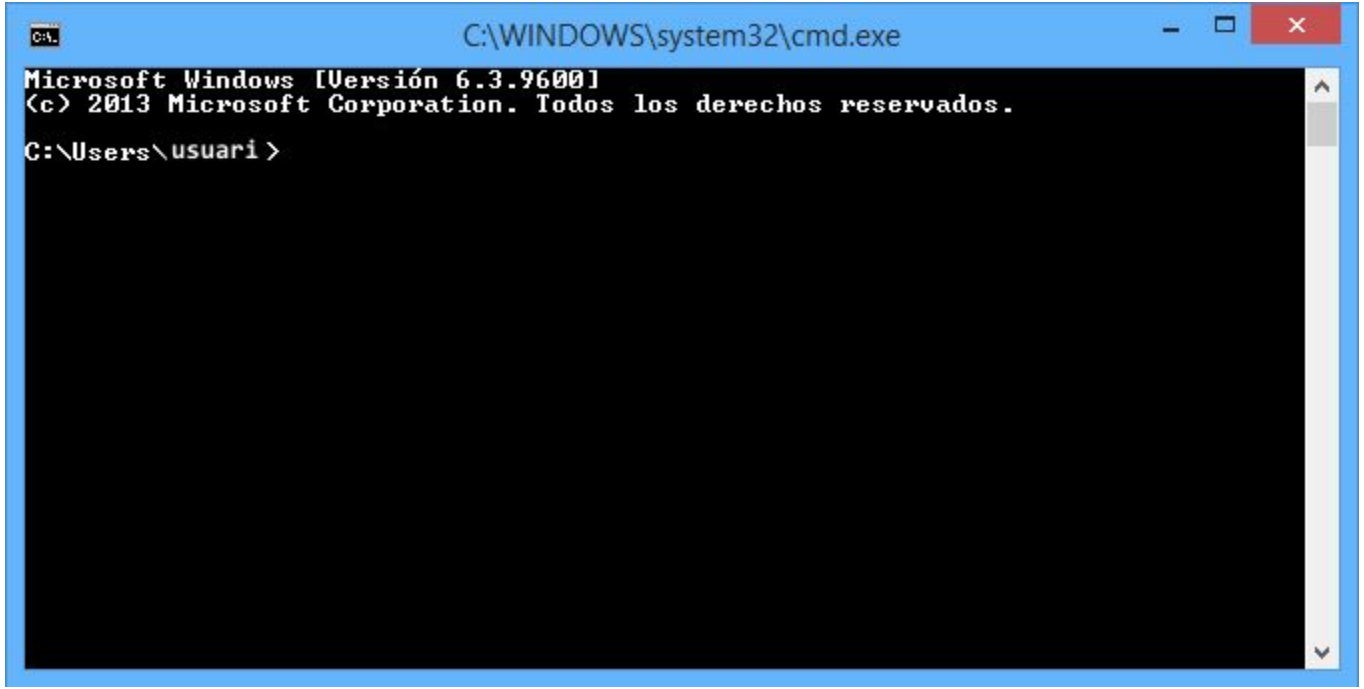
Only available for the recent Windows versions



Task Manager - overview of all processes running



Recycle Bin - temporary storage of deleted files



Command Prompt - will be explained shortly



```
PS C:\> Get-ChildItem 'MediaCenter:\Music' -rec |
>>         where { -not $_.PSIsContainer -and $_.Extension -match 'wma|mp3' } |
>>         Measure-Object -property length -sum -min -max -ave
>>


Count    : 1307
Average  : 5491276.09563887
Sum      : 7177097857
Maximum  : 22905267
Minimum  : 3235
Property : Length


PS C:\> Get-WmiObject CIM_BIOSElement | select biosv*, man*, ser* | Format-List


BIOSVersion  : {TOSCPL - 6040000, Ver 1.00PARTTBL}
Manufacturer : TOSHIBA
SerialNumber : M821116H



PS C:\> ([wmiSearcher]@'
>> SELECT * FROM CIM_Job
>> WHERE Priority > 1
>> '@).get() | Format-Custom
>>

class ManagementObject#root\cimv2\Win32_PrintJob
{
  Document = Monad Manifesto - Public
  JobId = 6
  JobStatus =
  Owner = User
  Priority = 42
  Size = 1027088
  Name = Epson Stylus COLOR 740 ESC/P 2, 6
}



PS C:\> $rssUrl = 'http://blogs.msdn.com/powershell/rss.aspx'
PS C:\> $blog = [xml](new-object System.Net.Webclient).DownloadString($rssUrl)
PS C:\> $blog.rss.channel.item | select title -first 3

title
-----
MMS: What's Coming In PowerShell V2
PowerShell Presence at MMS
MMS Talk:  System Center Foundation Technologies

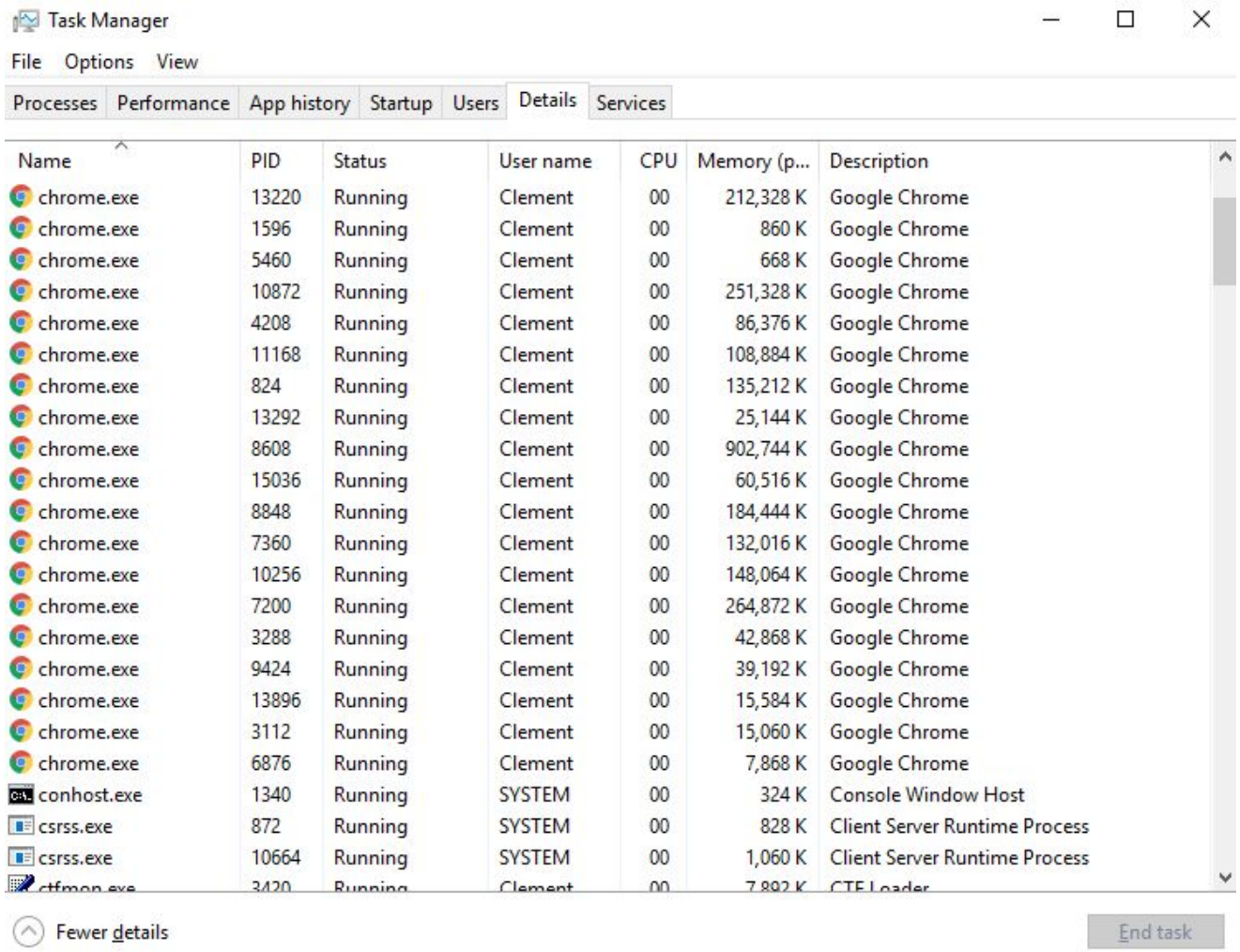

PS C:\> $host.version.ToString().Insert(0, 'Windows PowerShell: ')
Windows PowerShell: 1.0.0.0
PS C:\>
```

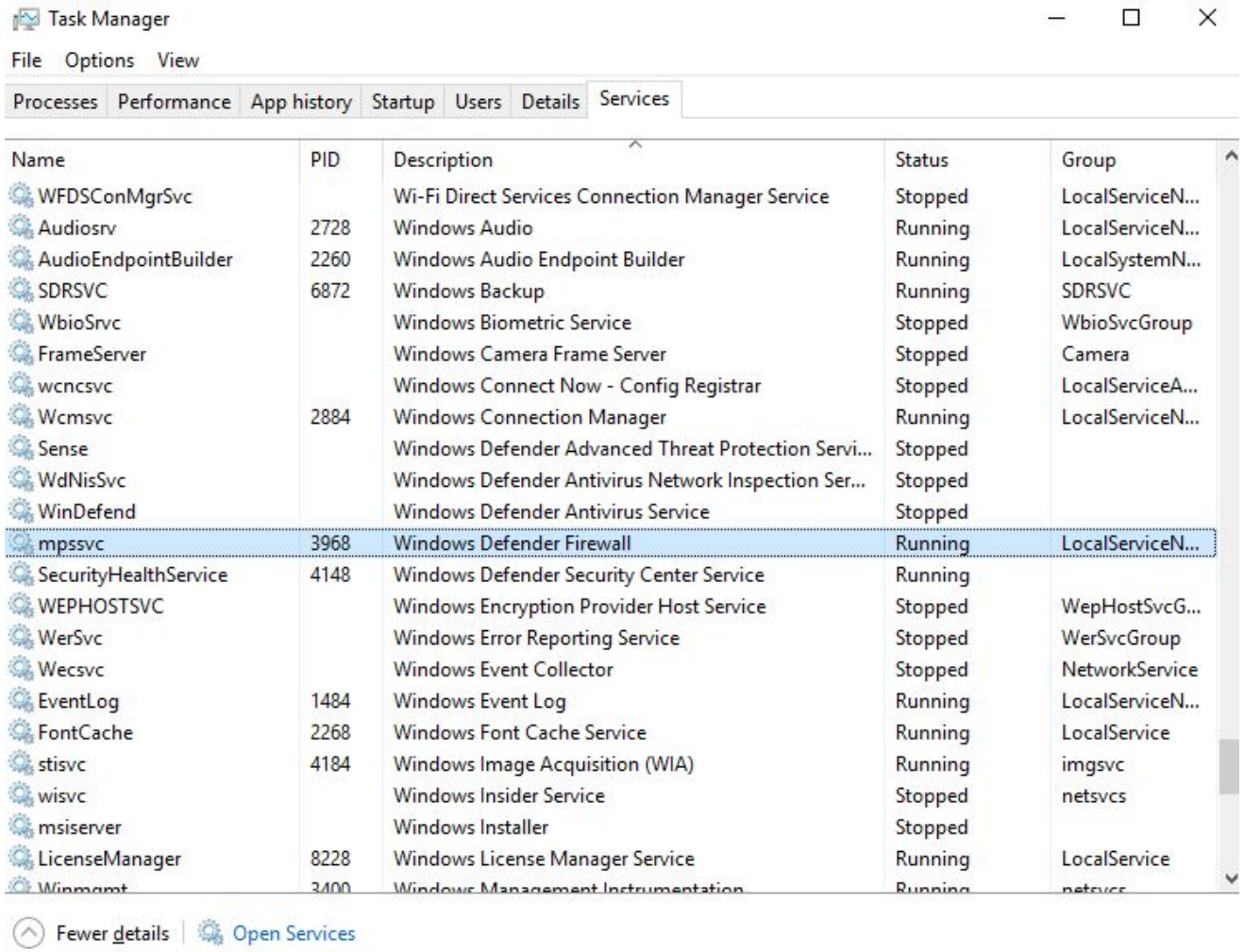
Powershell - will be explained shortly

More on Applications

An **application** may use multiple **processes** that perform a specific action, and **services** run in the background.

A web browser application may have one process for browser connection and another process for displaying the content, as an example.


Task Manager
File Options View
Processes Performance App history Startup Users Details Services
Name PID Status User name CPU Memory (p... Description
chrome.exe 13220 Running Clement 00 212,328 K Google Chrome
chrome.exe 1596 Running Clement 00 860 K Google Chrome
chrome.exe 5460 Running Clement 00 668 K Google Chrome
chrome.exe 10872 Running Clement 00 251,328 K Google Chrome
chrome.exe 4208 Running Clement 00 86,376 K Google Chrome
chrome.exe 11168 Running Clement 00 108,884 K Google Chrome
chrome.exe 824 Running Clement 00 135,212 K Google Chrome
chrome.exe 13292 Running Clement 00 25,144 K Google Chrome
chrome.exe 8608 Running Clement 00 902,744 K Google Chrome
chrome.exe 15036 Running Clement 00 60,516 K Google Chrome
chrome.exe 8848 Running Clement 00 184,444 K Google Chrome
chrome.exe 7360 Running Clement 00 132,016 K Google Chrome
chrome.exe 10256 Running Clement 00 148,064 K Google Chrome
chrome.exe 7200 Running Clement 00 264,872 K Google Chrome
chrome.exe 3288 Running Clement 00 42,868 K Google Chrome
chrome.exe 9424 Running Clement 00 39,192 K Google Chrome
chrome.exe 13896 Running Clement 00 15,584 K Google Chrome
chrome.exe 3112 Running Clement 00 15,060 K Google Chrome
chrome.exe 6876 Running Clement 00 7,868 K Google Chrome
conhost.exe 1340 Running SYSTEM 00 324 K Console Window Host
csrss.exe 872 Running SYSTEM 00 828 K Client Server Runtime Process
csrss.exe 10664 Running SYSTEM 00 1,060 K Client Server Runtime Process
Fewer details
End task


Task Manager
File Options View
Processes Performance App history Startup Users Details Services
Name PID Description Status Group
WFDSConMgrSvc Wi-Fi Direct Services Connection Manager Service Stopped LocalServiceN...
Audiosrv 2728 Windows Audio Running LocalServiceN...
AudioEndpointBuilder 2260 Windows Audio Endpoint Builder Running LocalSystemN...
SDRSVC 6872 Windows Backup Running SDRSVC
WbioSrvc Windows Biometric Service Stopped WbioSvcGroup
FrameServer Windows Camera Frame Server Stopped Camera
wcncsvc Windows Connect Now - Config Registrar Stopped LocalServiceA...
Wcmsvc 2884 Windows Connection Manager Running LocalServiceN...
Sense Windows Defender Advanced Threat Protection Servi... Stopped
WdNisSvc Windows Defender Antivirus Network Inspection Ser... Stopped
WinDefend Windows Defender Antivirus Service Stopped
mpssvc 3968 Windows Defender Firewall Running LocalServiceN...
SecurityHealthService 4148 Windows Defender Security Center Service Running
WEPHOSTSVC Windows Encryption Provider Host Service Stopped WepHostSvcG...
WerSvc Windows Error Reporting Service Stopped WerSvcGroup
Wecsvc Windows Event Collector Stopped NetworkService
EventLog 1484 Windows Event Log Running LocalServiceN...
FontCache 2268 Windows Font Cache Service Running LocalService
stisvc 4184 Windows Image Acquisition (WIA) Running imgsvc
wisvc Windows Insider Service Stopped netsvcs
msiserver Windows Installer Stopped
LicenseManager 8228 Windows License Manager Service Running LocalService
Fewer details
Open Services

Command Prompt

Do everything from one spot, with only commands

- Only recognizes a specific set of commands
- Some command may require you to be elevated as an administrator

CMD

Try running command prompt as administrator

You should see “C:\WINDOWS\system32>” if you are running as administrator (else, “C:\Users\<your user>”)

Try running some simple commands (next slide)

Some CMD commands

dir	Display all files and folders in the current directory
cd [path]	Change directory; specify the folder you want to enter, or use “cd ..” to go back one
echo [message]	Output a message to the command prompt
exit	Exits the command prompt
help	Print a list of some sample commands

Powershell

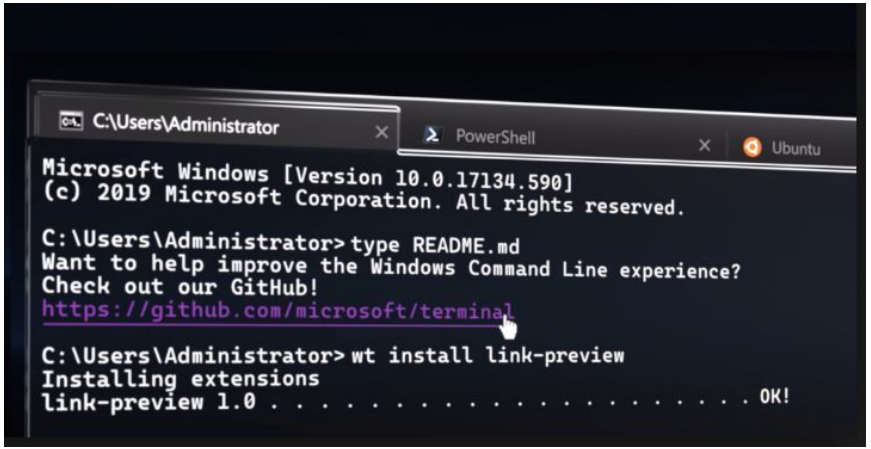


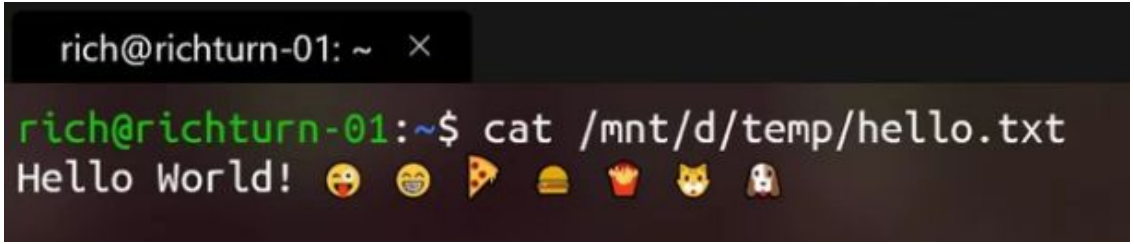
Even more powerful than CMD

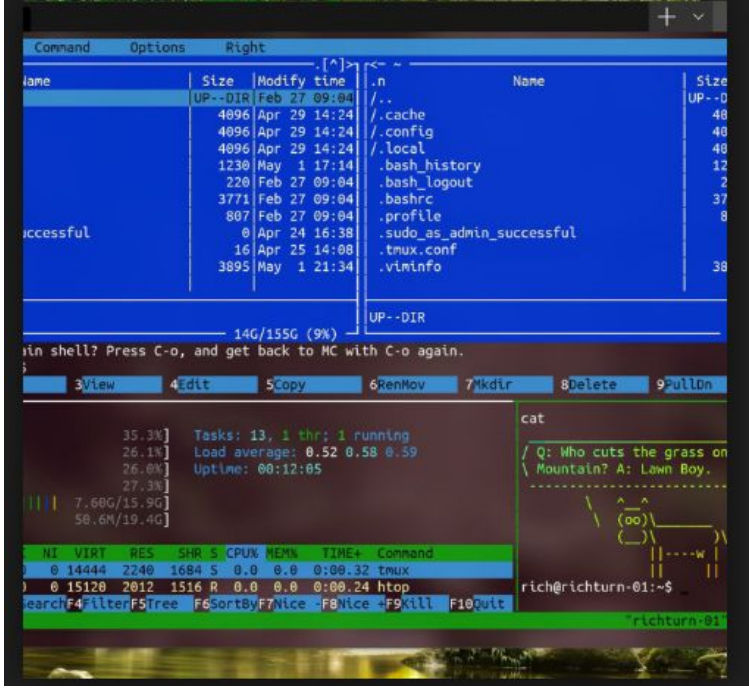
- Like cmd, only recognizes a specific set of commands (but there are many)
- Like cmd, some commands require elevation

Some differences

CMD	Powershell
Black and white by default	Blue and white by default
Has less commands	Includes commands from Linux (aliased commands)
Reliable	May not be installed (rarely)
Cannot expand its functionality	Can import modules for further extension
Harder to control output	Has colorful, beautiful formatting
Generally harder to use	Generally easier to use
For good old nostalgia	The new, modern look







There's also an upcoming newer version called
"*Windows Terminal*"
You can customize how it looks, you can make profiles
There's also support for ***emojis*** much wow

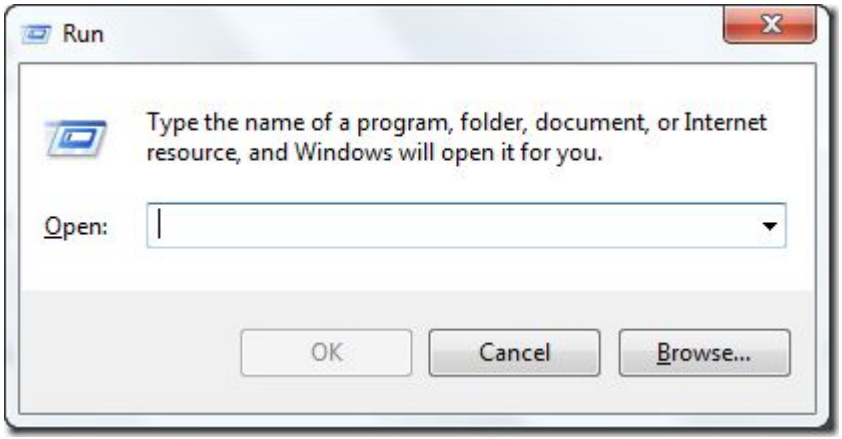
Buttons

Windows button

A crucial key that is commonly found on the bottom left of some keyboards, designated with the Windows logo

Increases efficiency





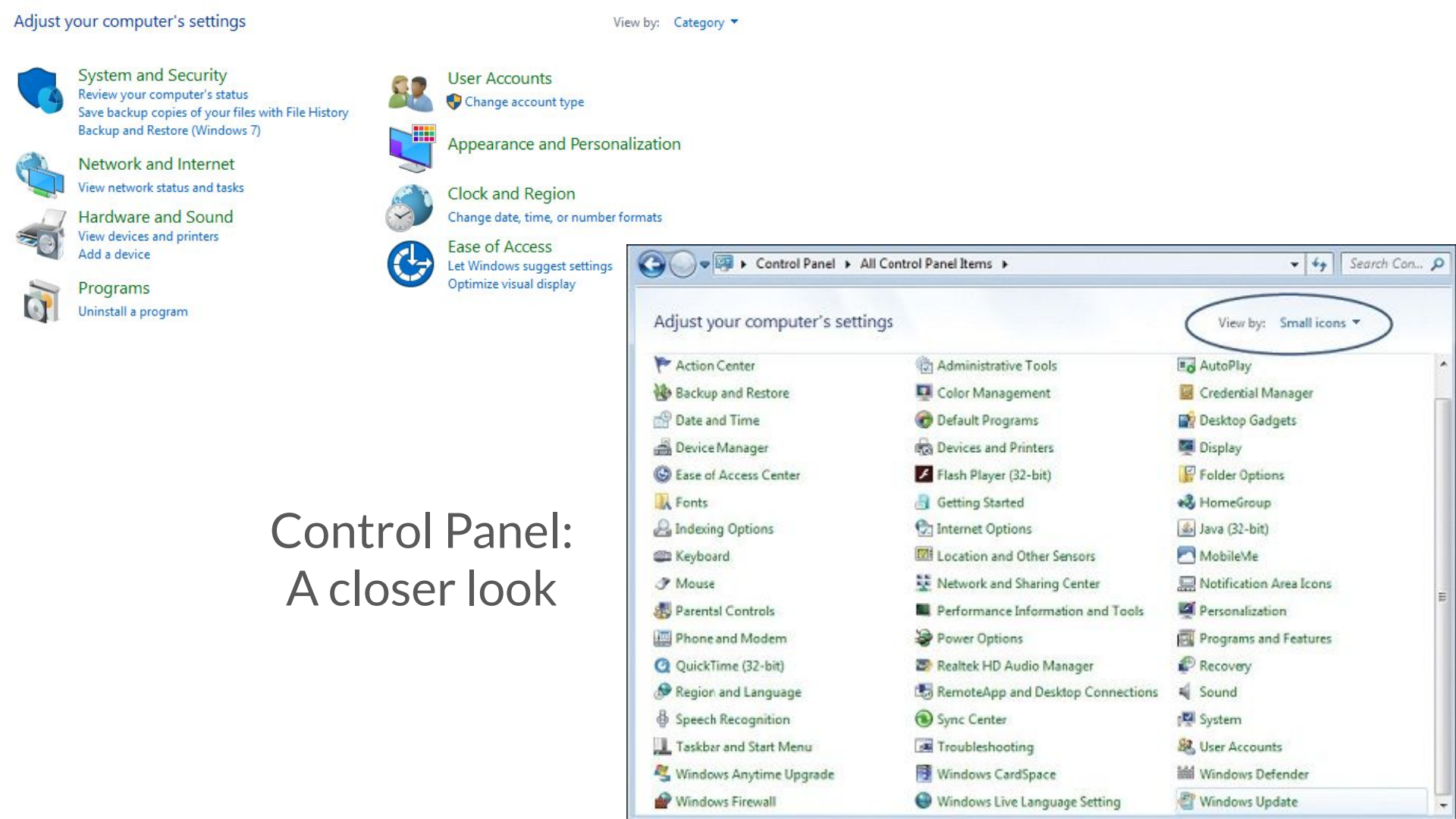
Win+R (run command) - used to frequently get somewhere or do something quickly
Try entering “cmd” or “chrome” in the prompt!

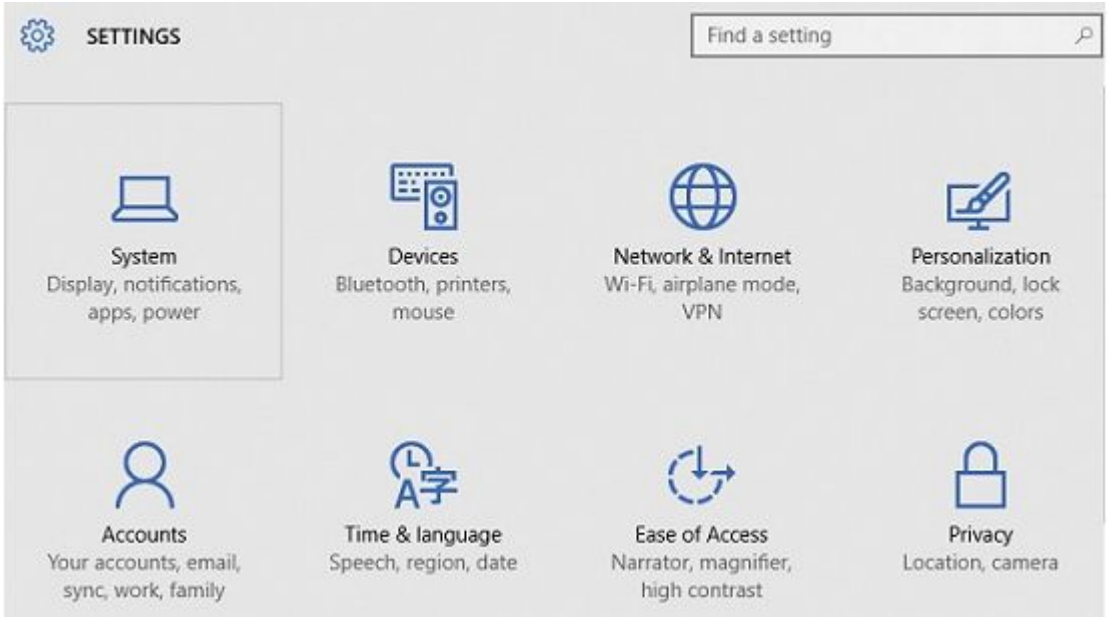
Control Panel

Control Panel

A method for configuring various components

Commonly used by some technicians


Adjust your computer's settings
View by: Category
System and Security
Review your computer's status
Save backup copies of your files with File History
Backup and Restore (Windows 7)
Network and Internet
View network status and tasks
Hardware and Sound
View devices and printers
Add a device
Programs
Uninstall a program
User Accounts
Change account type
Appearance and Personalization
Clock and Region
Change date, time, or number formats
Ease of Access
Let Windows suggest settings
Optimize visual display
Control Panel: A closer look
Control Panel ▸ All Control Panel Items ▸
Search Con...
Adjust your computer's settings
View by: Small icons
Action Center
Administrative Tools
AutoPlay
Backup and Restore
Color Management
Credential Manager
Date and Time
Default Programs
Desktop Gadgets
Device Manager
Devices and Printers
Display
Ease of Access Center
Flash Player (32-bit)
Folder Options
Fonts
Getting Started
HomeGroup
Indexing Options
Internet Options
Java (32-bit)
Keyboard
Location and Other Sensors
MobileMe
Mouse
Network and Sharing Center
Notification Area Icons
Parental Controls
Performance Information and Tools
Personalization
Phone and Modem
Power Options
Programs and Features
QuickTime (32-bit)
Realtek HD Audio Manager
Recovery
Region and Language
RemoteApp and Desktop Connections
Sound
Speech Recognition
Sync Center
System
Taskbar and Start Menu
Troubleshooting
User Accounts
Windows Anytime Upgrade
Windows CardSpace
Windows Defender
Windows Firewall
Windows Live Language Setting
Windows Update



Windows 8 and above include *Settings*, which is similar

Defender

Defender

Microsoft Vista, 7, 8, and 10 come this, which works with Internet Explorer/Edge to warn for spyware

All Windows Defender versions are trash except **Windows 10**'s which is actually pretty good

It's still wise to install an accompanying third-party antivirus software

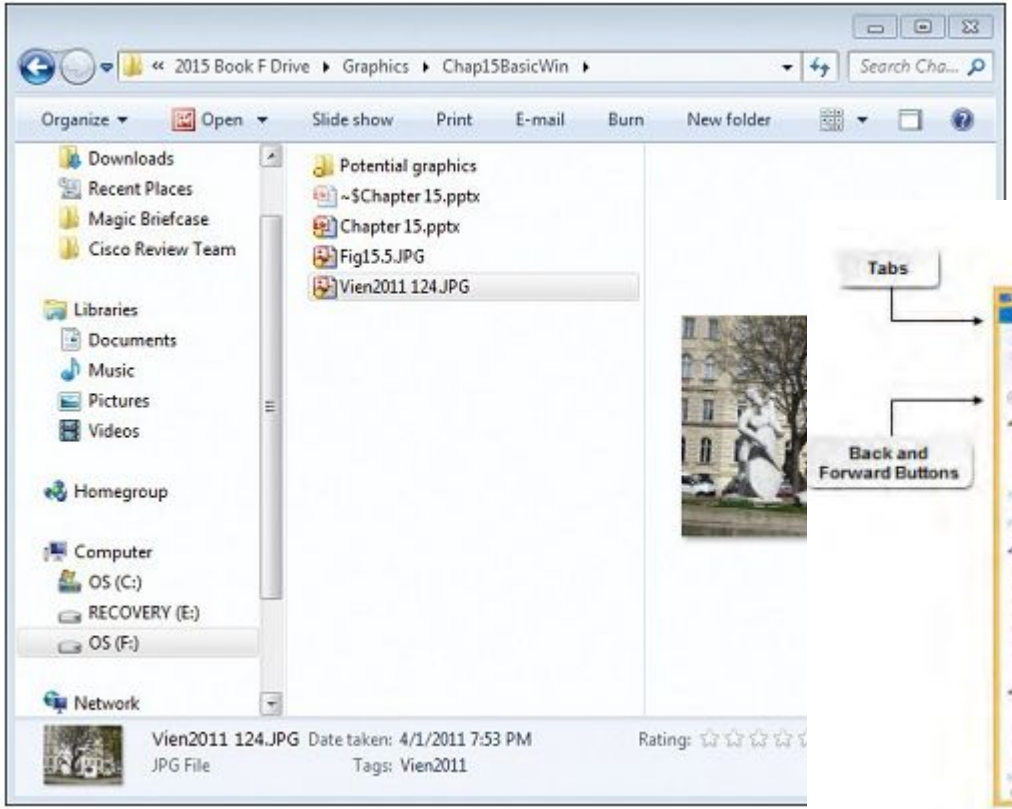
Explorer

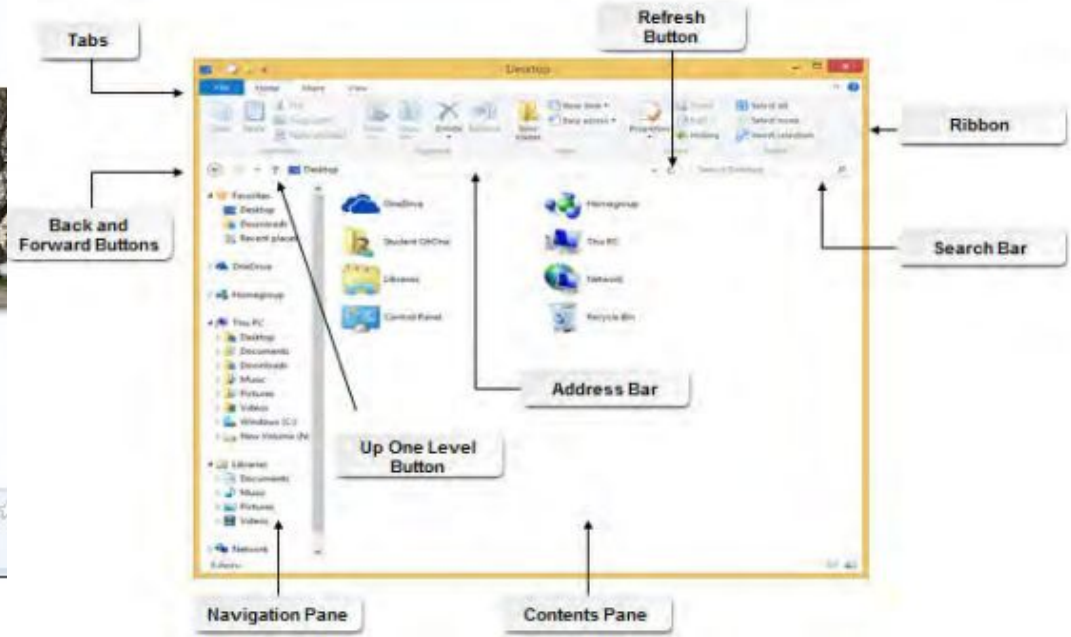
Explorer

Windows Explorer/File Explorer are used to manage files and folders

In the past, it was called Windows Explorer

Recently, it is now called File Explorer


2015 Book F Drive
Graphics
Chap15BasicWin
Search Cha...
Organize
Open
Slide show
Print
E-mail
Burn
New folder
Downloads
Recent Places
Magic Briefcase
Cisco Review Team
Libraries
Documents
Music
Pictures
Videos
Homegroup
Computer
OS (C:)
RECOVERY (E:)
OS (F:)
Network
Potential graphics
~$Chapter 15.pptx
Chapter 15.pptx
Fig15.5.JPG
Vien2011 124.JPG
Vien2011 124.JPG Date taken: 4/1/2011 7:53 PM
JPG File
Tags: Vien2011
Rating:


Tabs
Refresh Button
Ribbon
Back and Forward Buttons
Search Bar
Address Bar
Up One Level Button
Navigation Pane
Contents Pane

File structure

A file is an electronic container that holds computer code or data

A folder holds files and can also contain other folders

You can create, copy, or move files or folders with Windows Explorer/File Explorer

File types

File extensions indicate a file's type, and they help determine what program is used to open a file

As an example, you should know that .mp3 files are audio files. Plain text files end in .txt

There are many different extensions out there

Firewall

Firewall

Different types

- Hardware device that protects an organization
- Software available through the operating system
- Third-party vendor designed to protect a particular device

Firewall Use

A firewall protects one or more computers from outside attacks

You can think of the firewall as a filter - it controls access to and from the computer

Graphical User Interface (GUI)

Graphical User Interface (GUI)

Form of user interface that allows users to interact with electronic devices through graphical icons, visual indicators, etc.

Not supposed to be text-based

Windows Desktop

Area on the screen of a GUI environment in which all work is performed

Contains icons, can be customized

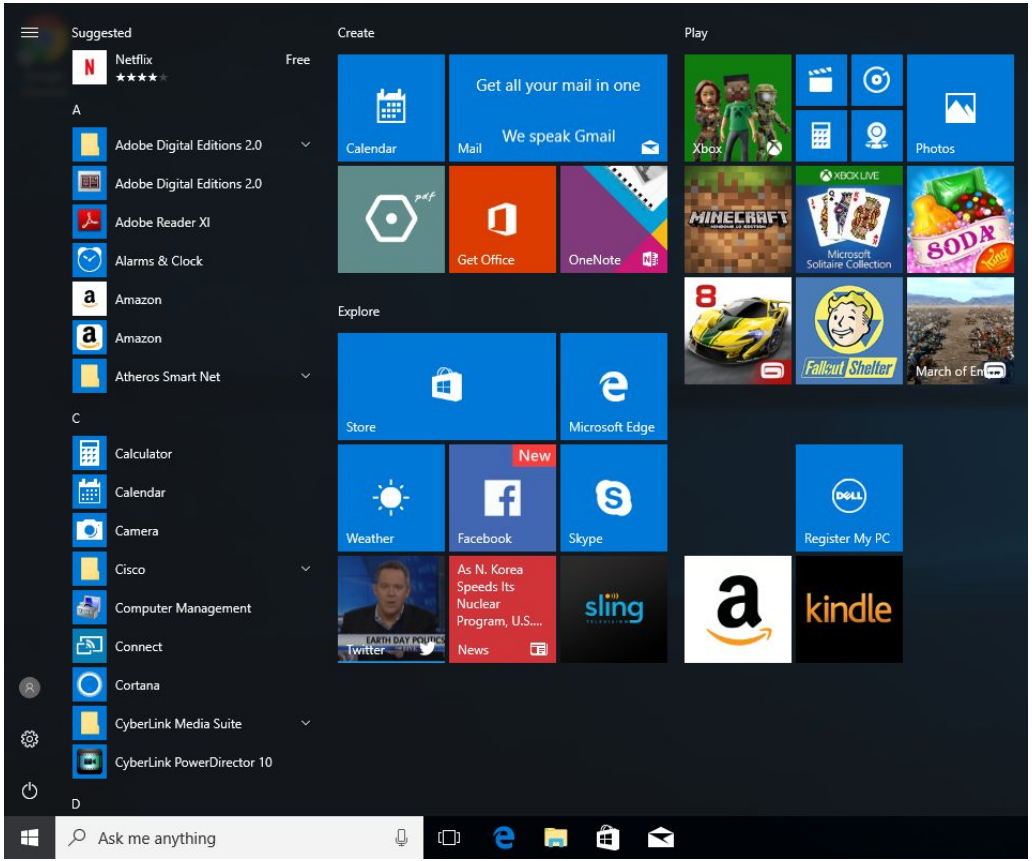


Windows Desktop example

Start Menu

Customizable and common central point for:

- Launching applications
- Opening recent documents
- Accessing system settings
- Finding files
- Getting help



Start Menu for Windows 10

Start Button

The Start button by default is located in the desktop's lower-left corner on the taskbar and is used to open the start menu

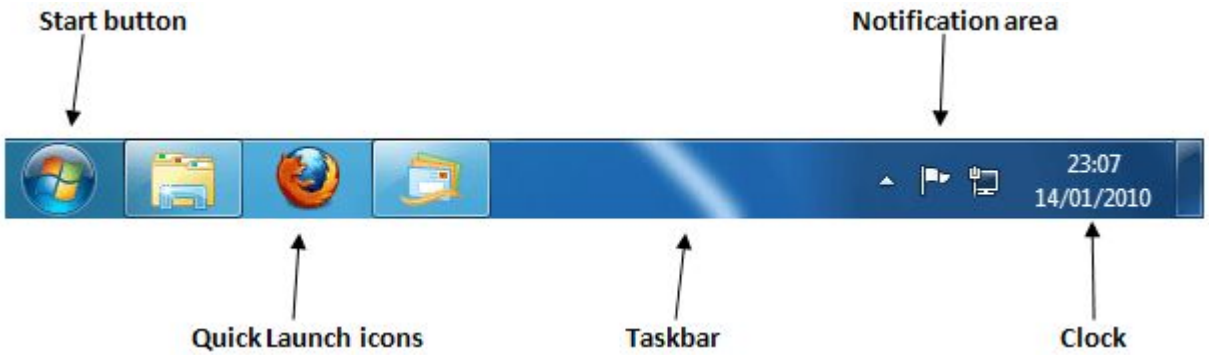
Start Menu

From here, you can also shutdown, restart, or do other things with your computer



Taskbar

Typically located at the bottom, it is used to launch and monitor applications



As a summary, make sure you know:

- Applications
- Buttons (Windows button)
- Control Panel
- Defender
- Explorer
- Firewall
- GUIs